EXPERIMENT-2

Aim :- Convert the ER diagram into a relational schema. Write complete MYSQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data demonstrate referential integrity violations.

Source Code :-

```
DROP DATABASE IF EXISTS ecommerce_db;
CREATE DATABASE ecommerce_db;
USE ecommerce_db;

-- 1. CUSTOMER (strong entity)

CREATE TABLE Customer(

        customer_id  INT  AUTO_INCREMENT PRIMARY KEY,
        first_name   VARCHAR(50) NOT NULL,
        last_name    VARCHAR(50) NOT NULL,
        email        VARCHAR(100) NOT NULL UNIQUE

) ENGINE = InnoDB;
```

```
-- Multi-valued attribute: Customer phone numbers
CREATE TABLE Customer_Phone (
        customer_id  INT NOT NULL,
        phone  VARCHAR(15) NOT NULL,
        PRIMARY KEY (customer_id, phone),
        FOREIGN KEY (customer_id) REFRENCES Customer(CustomerId)
                ON DELETE CASCADE
) ENGINE = InnoDB;

-- 2. ADDRESS (weak entity - depends on Customer)
CREATE TABLE Address (
        customer_id  INT NOT NULL,
        address_id   INT NOT NULL,    -- partial key
        street       VARCHAR(100) NOT NULL,
        city         VARCHAR(50) NOT NULL,
        State        VARCHAR(50)  NOT NULL,
        pincode      VARCHAR(10)  NOT NULL,
        PRIMARY KEY (customer_id, address_id),   -- composite
        FOREIGN KEY (customer_id) REFRENCES      KEY of weak
                Customer(Customer_id) ON DELETE CASCADE
                                                 entity
);

) ENGINE = InnoDB;
```

```
--3. CATEGORY (strong entity)
CREATE TABLE Category (
        Category_id   INT AUTO_INCREMENT PRIMARY KEY,
        Category_name   VARCHAR(50) NOT NULL UNIQUE
) ENGINE = InnoDB;

--4. SELLER (strong entity)
CREATE TABLE Seller (
        seller_id   INT  AUTO_INCREMENT PRIMARY KEY,
        seller_name   VARCHAR(100) NOT NULL,
        email     VARCHAR(100) NOT NULL UNIQUE
) ENGINE = InnoDB;

CREATE TABLE Seller_Phone (
        seller_id   INT NOT NULL,
        phone     VARCHAR(15) NOT NULL,
        PRIMARY KEY (seller_id, phone),
        FOREIGN KEY (seller_id) REFERENCES Seller(seller_id)
                ON DELETE CASCADE
) ENGINE = InnoDB;
```

```
--5. PRODUCT (strong entity, with specialization)
CREATE TABLE Product (
    product_id   INT AUTO_INCREMENT PRIMARY KEY,
    product_name   VARCHAR(150) NOT NULL,
    price          DECIMAL(10,2) NOT NULL,
    stock          INT NOT NULL DEFAULT 0,
    category_id    INT NULL,
    seller_id      INT NOT NULL,
    FOREIGN KEY(category_id) REFRENCES Category(category_id)
            ON DELETE SET NULL,
    FOREIGN KEY (seller_id) REFRENCES Seller(seller_id)
            ON DELETE CASCADE
) ENGINE = InnoDB;

-- Specialization: Electronics (disjoint subtype of Product)
CREATE TABLE Electronics (
        product_id  INT PRIMARY KEY,
        warranty_period INT,          -- in months
        brand VARCHAR(50),
        FOREIGN KEY (product_id) REFRENCES Product(product_id)
            ON DELETE CASCADE
) ENGINE = InnoDB;
```

```
--specialization: Clothing (disjoint subtype of Product)
CREATE TABLE Clothing (
        product_id INT PRIMARY KEY,
        Size VARCHAR(10),
        fabric VARCHAR(50),
        FOREIGN KEY (product_id) REFRENCES Product
          (product_id)
                ON DELETE CASCADE
) ENGINE = InnoDB;

--6. ORDER (strong Entity)- "Order" is a reserved word in MySQL
CREATE TABLE Orders (
        Order_id INT AUTO_INCREMENT PRIMARY KEY,
        order_date DATE NOT NULL,
        total_amount DECIMAL (10,2) NOT NULL,
        customer_id INT NOT NULL,
        FOREIGN KEY (Customer_id) REFERENCES Customer (customer_id)
                ON DELETE CASCADE

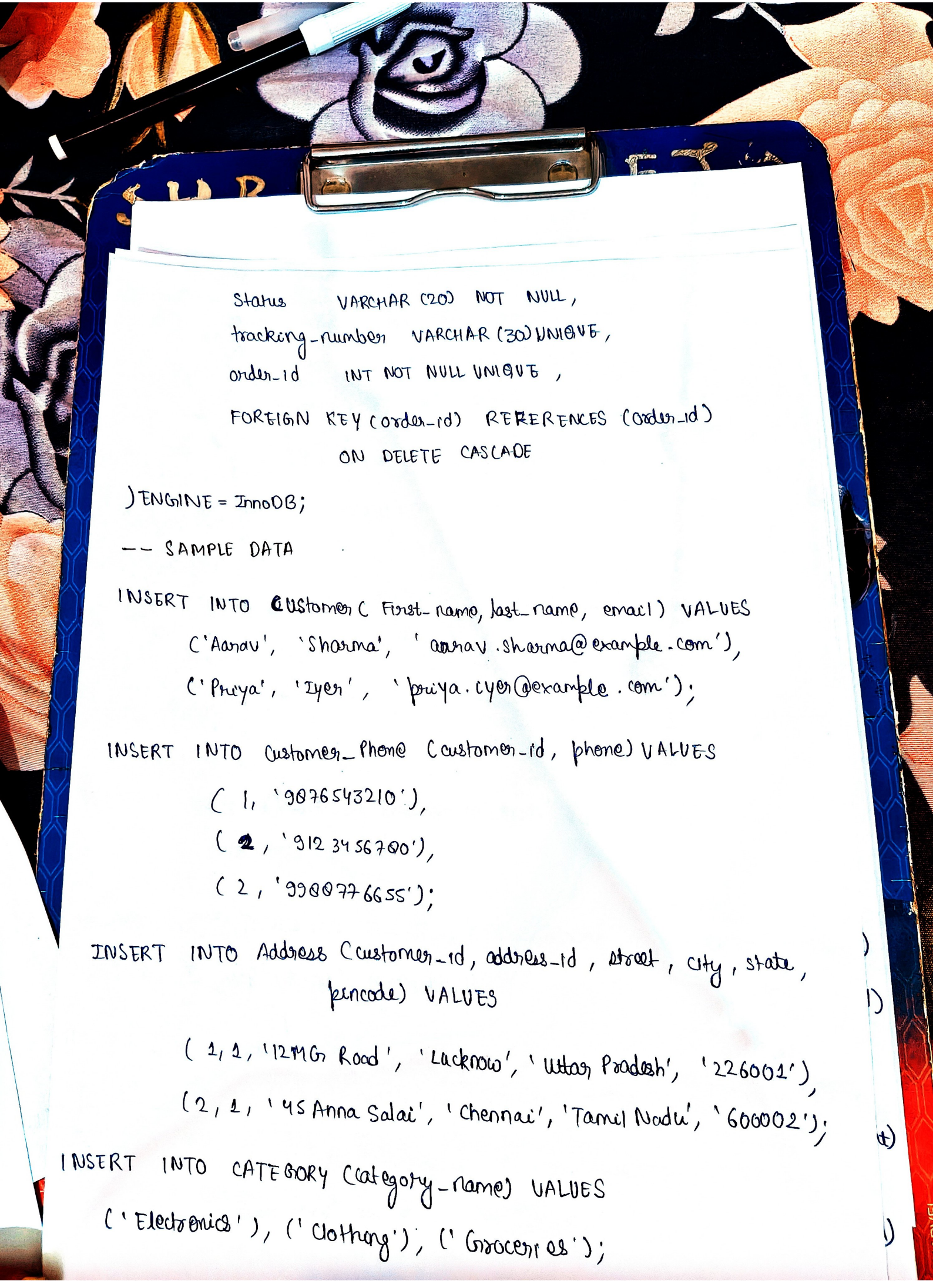
) ENGINE = InnoDB;

--7. ORDER_ITEM (weak entity - depends on Orders)
CREATE TABLE Order_Item (
        Order_id INT NOT NULL,
        item_no INT NOT NULL,      --partial key
```

```
    quantity      INT NOT NULL,
    unit_price    DECIMAL(10,2) NOT NULL,
    product_id    INT NULL,
    PRIMARY KEY (order_id, item_no), -- composite key of weak entity
    FOREIGN KEY (order_id) REFERENCES Orders(order_id)
        ON DELETE CASCADE,
    FOREIGN KEY (product_id) REFERENCES Product
        (product_id) ON DELETE SET NULL
) ENGINE = InnoDB;

-- 8. PAYMENT (strong entity, 1:1 with Orders)
CREATE TABLE Payment(
    payment_id    INT AUTO_INCREMENT PRIMARY KEY,
    amount        DECIMAL(10,2) NOT NULL,
    payment_date  DATE NOT NULL,
    payment_mode  VARCHAR(20) NOT NULL,
    order_id      INT NOT NULL UNIQUE, -- UNIQUE enforces the 1:1 cardinality
    FOREIGN KEY (order_id) REFERENCES Orders(order_id)
        ON DELETE CASCADE
) ENGINE = InnoDB;

-- 9. DELIVERY (strong entity, 1:1 with Orders)
CREATE TABLE Delivery(
    delivery_id   INT AUTO_INCREMENT PRIMARY KEY,
    delivery_date DATE,
```



```
    Status          VARCHAR(20) NOT NULL,
    tracking_number VARCHAR(30) UNIQUE,
    order_id        INT NOT NULL UNIQUE,
    FOREIGN KEY(order_id) REFERENCES (order_id)
        ON DELETE CASCADE
)ENGINE = InnoDB;

-- SAMPLE DATA

INSERT INTO Customer( First_name, last_name, email) VALUES
    ('Aarav', 'Sharma', 'aarav.sharma@example.com'),
    ('Priya', 'Iyer', 'priya.iyer@example.com');

INSERT INTO Customer_Phone (customer_id, phone) VALUES
    (1, '9876543210'),
    (2, '9123456700'),
    (2, '9900776655');

INSERT INTO Address (customer_id, address_id, street, city, state,
    pincode) VALUES

    (1, 1, '12MG Road', 'Lucknow', 'Uttar Pradesh', '226001'),
    (2, 1, '45 Anna Salai', 'Chennai', 'Tamil Nadu', '600002');

INSERT INTO CATEGORY (category_name) VALUES
('Electronics'), ('Clothing'), ('Groceries');
```

1a)

```
INSERT INTO Seller (seller_name, email) VALUES
('TechBazar Pvt Ltd', 'contact@techbazar.com'),
('FashionHub India', 'support@fashionhub.in');

INSERT INTO Seller_Phone (Seller_id, phone) VALUES
(1, '011-40001234'),
(2, '022-50005678');

INSERT INTO Product (product_name, price, stock, category_id,
seller_id) VALUES
('Wireless Earbuds X200', 2499.00, 50, 1, 1),
('Cotton Kurta - Blue', 899.00, 100, 2, 2);

INSERT INTO Clothing (product_id, size, fabric) VALUES
(2, 'M', 'cotton');

INSERT INTO Orders (order_date, total_amount, customer_id) VALUES
('2026-08-01', 3398.00, 1);

INSERT INTO Electronics (product_id, warranty_period, brand) VALUES
(1, 12, 'Soundwave');

INSERT INTO order_Item (order_id, item_no, quantity, unit_price,
product_id) VALUES
(1, 1, 1, 2499.00, 1),
(1, 2, 1, 899.00, 2);
```

```
INSERT INTO Payment (amount, payment_date, payment_mode, order_id)
    VALUES (3388.00, '2026-08-01', 'UPI', 1);

INSERT INTO Delivery (delivery_date, status, tracking_number,
    order_id) VALUES
('2026-08-05', 'Shipped', 'TRK123456789', 1);

-- DEMONSTRATING REFERENTIAL INTEGRITY
-(a) INSERT violation: order_id 999 does not exist in Orders
-- Expected: Error 1452 - Cannot add or update a child row:
    a foreign key constraint fails (order_id FK)

-- INSERT INTO Order_Item (order_id, item_no, quantity, unit_price,
    product_id)

-- VALUES (999, 1, 2, 500.00, 1);

-- (b) INSERT violation: product_id 999 does not exist in Product
-- Expected: Error 1452 - foreign key constraint fails (product_id FK)
-INSERT INTO Order_Item (order_id, item_no, quantity, unit_price, product_id)
-- VALUES (1, 3, 1, 100.00, 999);

-- (c) UNIQUE violation: this order already has a payment (1:1 constraint)
-- Expected: Error 1062- Duplicate entry '1' for key 'order_id'
-- INSERT INTO Payment (amount, payment_date, payment_mode, order_id)
-- Values (500.00, '2026-08-02', 'Cash', 1);
```



```
-- (d) ON DELETE SET NULL demonstration: delete a category
-- Expected: product.category_id for its products becomes NULL, rows
   survive
-- Delete FROM category WHERE category_id = 1;
-- SELECT * FROM Product WHERE product_id = 1 ; -- category_id is now
                                                   NULL


-- (e) ON DELETE CASCADE demonstration : delete a customer
-- Expected: their Address, Customer_Phone, Orders rows are deleted
--          automatically, which in turn cascades to order_item,
--          Payment, and Delivery for for those orders

-- DELETE From Customer WHERE customer_id = 1;
-- SELECT * FROM Orders WHERE customer_id = 1; -- empty result
-- SELECT * FROM Payment WHERE order_id = 1 ; -- empty result

-- (f) NOT NULL violation: order_date is required
-- Expected: Error 1364- Field 'order_date' doesn't have a default
                                                   value.
-- INSERT INTO orders ( total_amount, customer_id) VALUES
                                                   (500.00, 1);
```